Project Initialization and planning phase

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2	Used to create a simple, responsive and user-friendly interface for entering fitness details and displaying workout plans.
2	Backend / Server-Side	Python, FastAPI	FastAPI handles user requests, API routes and communication between the frontend, database and Gemini AI.
3	Database / Data Storage	SQLite, SQL Alchemy	SQLite stores user details and workout plans, while SQL Alchemy provides easy database management using Python.
4	Cloud / Hosting / Deployment	Unicorn (Local Deployment)	Unicorn runs the Fast API application efficiently during development and testing.
5	Version Control & CI/CD	Git, GitHub	Used for version control, project backup, code management and collaboration.
6	Third-Party APIs / Other Tools	Google Gemini API	Gemini AI generates personalized 7-day workout plans and nutrition/recovery suggestions based on user inputs.

Project Proposal (Proposed Solution) Report

The proposal aims to provide personalized fitness planning using Generative AI. Fit Buddy addresses the difficulty of creating suitable workout plans by generating customized 7-day workout plans based on user information and fitness goals. It also provides nutrition and recovery suggestions and allows users to update plans through feedback

Project Overview

Objective:

The primary objective is to develop an AI-powered fitness planning application that generates personalized and structured workout plans using Google Gemini models.

Scope:

The project provides personalized 7-day workout plans based on user details such as age, weight, fitness goal and workout intensity. It also includes nutrition/recovery tips and feedback-based plan updates.

Problem Statement

Description:

Many users find it difficult to create a suitable and structured workout plan based on their individual fitness goals and requirements.

Impact:

Solving this problem can provide users with a simple, personalized and structured fitness planning experience, helping them follow an appropriate weekly workout routine.

Proposed Solution

Approach:

FitBuddy uses Google Gemini AI models to analyze user-provided fitness information and generate a personalized 7-day workout plan. Users can also provide feedback to update and improve their generated plan.

Key Features:

- AI-based personalized 7-day workout plan generation.
- Workout plans based on age, weight, fitness goal and intensity.
- Nutrition and recovery suggestions.
- Feedback-based workout plan updates.
- User-friendly web interface.
- Storage of user details and workout plans.

Resource Requirements

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